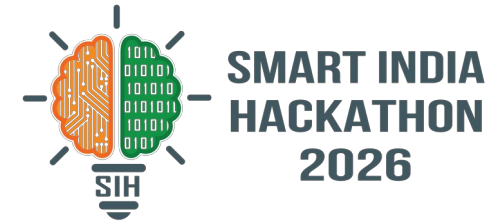
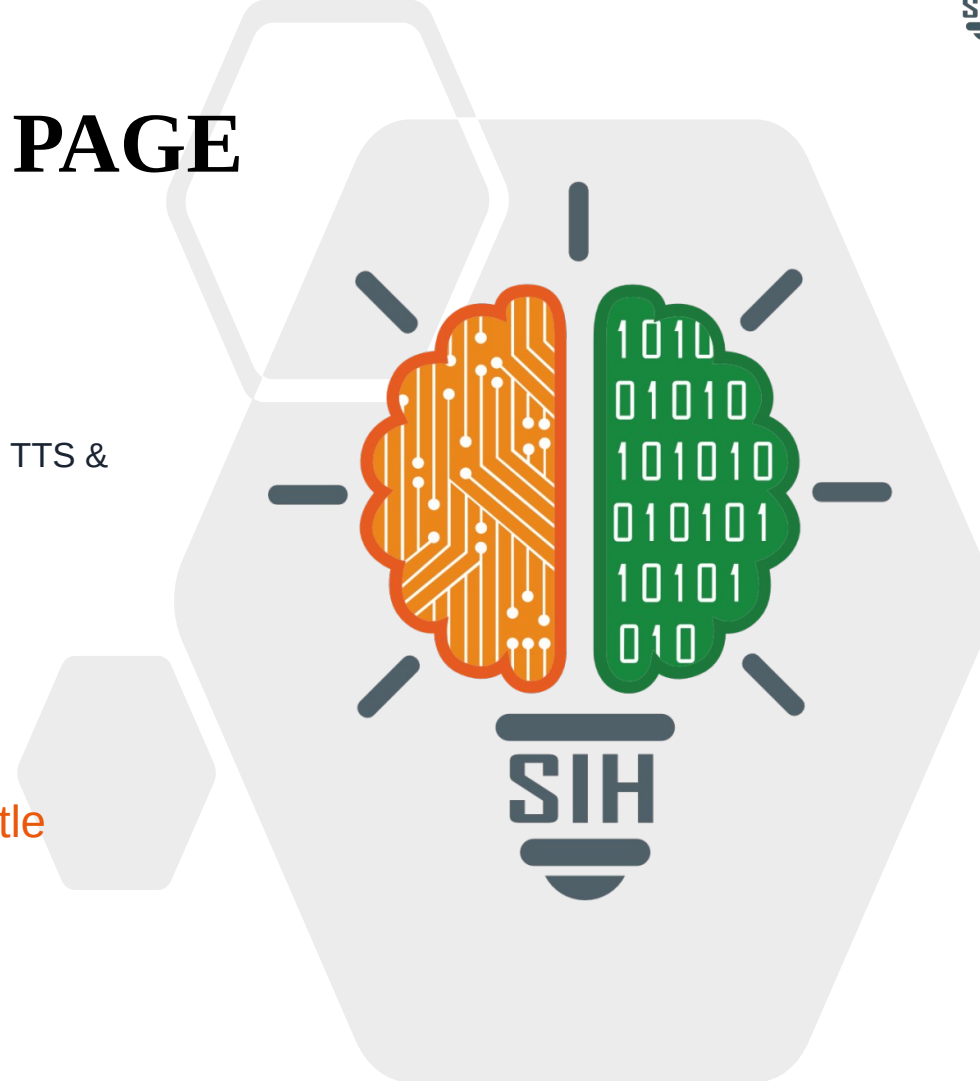


# SMART INDIA HACKATHON 2026



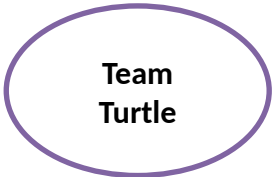
## TITLE PAGE

- **Problem Statement ID** – SIH26173
- **Problem Statement Title**- iTantra – Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- **Theme**- Smart Automation
- **PS Category**- Software
- **Team ID**- 188881
- **Team Name (Registered on portal)** Team Turtle

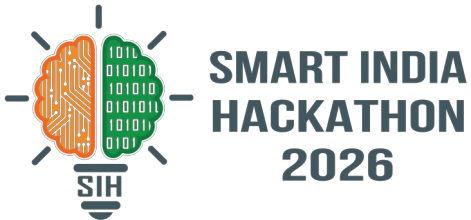


**iTantra**

Speak. Send text. Hear it back — offline.



# iTantra: Speak. Send Text. Hear It Back — Offline

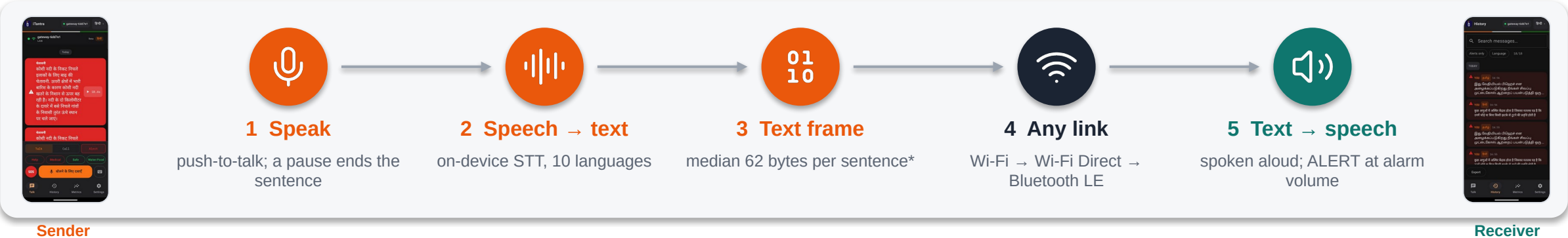


## Proposed Solution (Describe your Idea/Solution/Prototype)

An Android app that turns speech into text on the sender's phone, sends only the text, and speaks it aloud on the receiver's phone. **No tower, no internet, no cloud.**

Demo film  
1 min 55 s  
[itantra-106.pages.dev/demo](https://itantra-106.pages.dev/demo)





### Detailed explanation of the proposed solution

-  10 Indian languages, speech and voice, fully on the phone
-  ALERT plays at alarm volume; one-tap SOS sends GPS/NavIC location
-  Push-to-talk walkie-talkie and hands-free call mode

### How it addresses the problem

| Format       | Size (bits) |
|--------------|-------------|
| Raw audio    | 256k        |
| Opus voice   | 12k         |
| iTantra text | 165         |

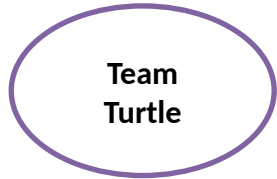
*bits per second, one spoken sentence (log scale)*

**Output is still speech** — people who cannot read are included

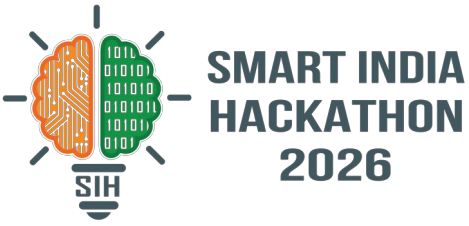
### Innovation and uniqueness of the solution

-  Speech → text → speech used as the codec for thin links
-  Alerts re-voiced on arrival, cannot be interrupted
-  Auto link fallback, one packet format on every link
-  LITE profile switches on by itself below 6 GB RAM

\*Median over 310 FLEURS sentences in a unit test; excludes encryption, alert signature and link headers. Bars assume a ~3 s sentence.



# TECHNICAL APPROACH



Technologies to be used (e.g. programming languages, frameworks, hardware)

**Android app**  
Kotlin · Jetpack Compose

**Voice detection**  
Silero VAD

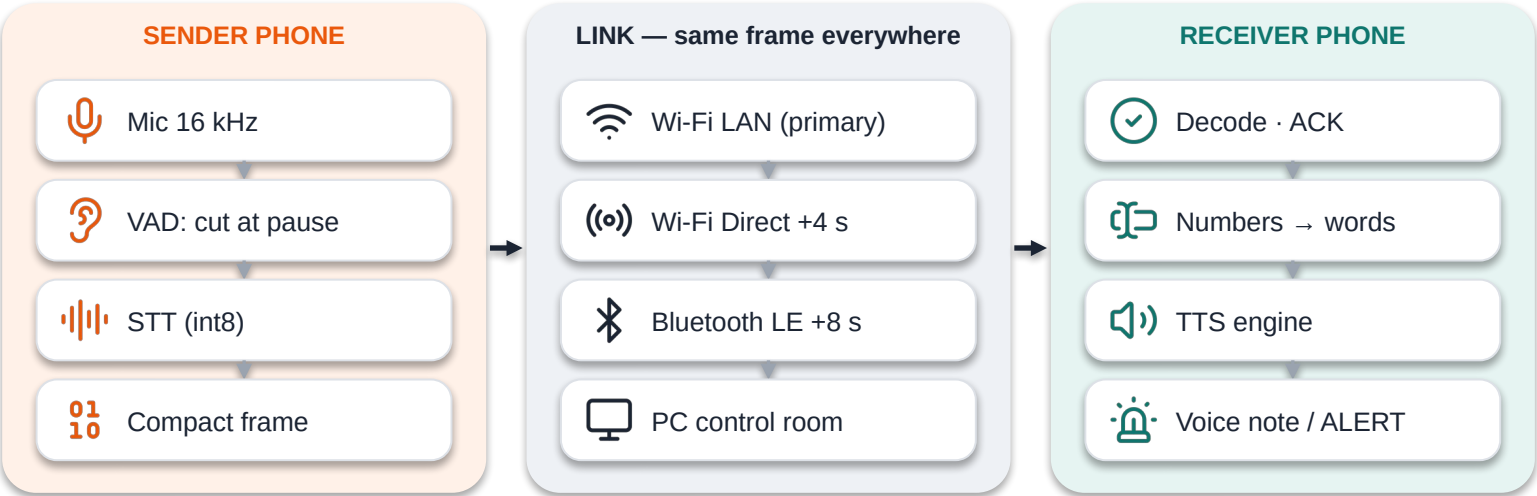
**Speech-to-text**  
SraVaani-1.0 · int8 · sherpa-onnx

**Text-to-speech**  
VITS Rasa · Indic-Mio

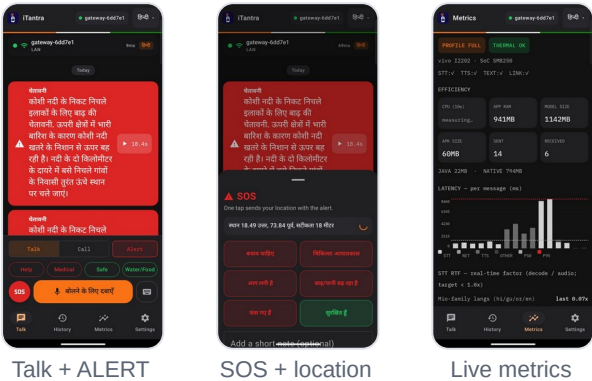
**Links**  
Wi-Fi (mDNS/TCP) · Wi-Fi Direct · BLE

**Hardware**  
Any Android phone · PC control room

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Working prototype — real screens



**BUILT & RUNNING**

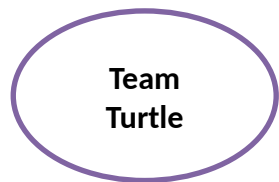
Push-to-talk, call mode, 10-language STT/TTS, ALERT, SOS + location, LITE profile, compact frame, PC control room

**BUILT · TEST PENDING**

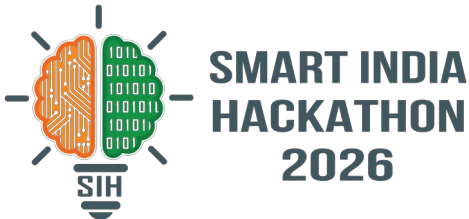
Wi-Fi Direct and Bluetooth LE fallback, phone-to-phone

**PLANNED**

Smaller voices (hi/gu/or/en), per-language model download, encrypted SOS location, ESP32 radio bridge



# FEASIBILITY AND VIABILITY



## Analysis of the feasibility of the idea

Measured in a 4 GB-class memory test (in-app logging)

**14×**  
faster than speech  
STT real-time factor 0.07

**0.6 s**  
speech end → text ready  
median, 10 languages

**765 MB**  
app RAM when idle  
1.8 GB peak while speaking

**0.3%**  
idle CPU  
of one core

**Alive**  
app not killed  
115 other apps closed

4 GB-class test: 12 GB Snapdragon 870 phone with RAM locked so only ~1.7 GB was free, LITE profile. The chip was not slowed — a budget phone will be slower. Word error rate 5.9% (preliminary, 10 clips) unchanged.

## Potential challenges and risks

## Strategies for overcoming these challenges

Voice starts 3–6 s after arrival in LITE

→ Keep the listener's language loaded; smaller fast voices for hi/gu/or/en (planned)

1.14 GB of model files

→ Per-language on-demand download and smaller voices for hi/gu/or/en (planned)

Wi-Fi Direct / Bluetooth not yet tested phone-to-phone

→ Two-phone tests on real devices; normal Wi-Fi stays the primary link

Noise, accents, code-mixing raise errors

→ Keyword boosting for alert words; 30-clip-per-language evaluation; typed text as fallback

## Viability

No airtime, server or cloud fees

Open models: MIT · Apache-2.0 · CC-BY-4.0

One packet format on every link

## Potential impact on the target audience



### Disaster response teams

Voice coordination when towers and internet are down



### Coastal fishing communities

Spoken local-language alerts over weak links



### Low-literacy users

Hear an alert a text message would miss



### Remote field staff

Offline voice link in poor-coverage areas

≈1,500×

less data than raw audio\*

1.03 s

ALERT arrival → spoken aloud  
(measured, Wi-Fi)

10

Indian languages, speech in and out, on the phone

## Benefits of the solution (social, economic, environmental, etc.)

### Social



Alerts spoken aloud in Indian languages — usable by people who cannot read

### Economic



No airtime, cloud or licence fees; runs on phones people already own

### Environmental



Fewer bytes per message means less radio airtime (battery gain not yet measured)

\*62 B per ~3 s sentence vs 256 kbps raw audio; 62 B is a unit-test median. No field pilot yet. Photos (Wikimedia Commons): NDRF – Ministry of Home Affairs, GODL-India; fishermen – Saral Shots, CC0; woman on phone – Ammalu, CC BY-SA 3.0; rail workers – David Brossard, CC BY-SA 2.0.

## Details / Links of the reference and research work



### Open models (all run on the phone)

SraVaani-1.0 speech-to-text · ARTPARK, IISc · MIT  
[huggingface.co/ARTPARK-IISc/SraVaani-1.0](https://huggingface.co/ARTPARK-IISc/SraVaani-1.0)

Indic-Mio text-to-speech · SPRING Lab, IIT Madras · Apache-2.0  
[huggingface.co/SPRINGLab/Indic-Mio](https://huggingface.co/SPRINGLab/Indic-Mio)

VITS Rasa 13 text-to-speech · AI4Bharat · CC-BY-4.0  
[huggingface.co/ai4bharat/vits\\_rasa\\_13](https://huggingface.co/ai4bharat/vits_rasa_13)

Silero VAD voice detection · MIT  
[github.com/snakers4/silero-vad](https://github.com/snakers4/silero-vad)



### Runtime and evaluation data

sherpa-onnx on-device ASR/TTS runtime · Apache-2.0  
[github.com/k2-fsa/sherpa-onnx](https://github.com/k2-fsa/sherpa-onnx)

Google FLEURS speech clips (STT evaluation)  
[huggingface.co/datasets/google/fleurs](https://huggingface.co/datasets/google/fleurs)



### Android platform guides

Wi-Fi Direct (Wi-Fi P2P)  
[developer.android.com/develop/connectivity/wifi/wifi-direct](https://developer.android.com/develop/connectivity/wifi/wifi-direct)

Bluetooth and BLE GATT  
[developer.android.com/develop/connectivity/bluetooth](https://developer.android.com/develop/connectivity/bluetooth)

Audio focus (ALERT playback)  
[developer.android.com/media/optimize/audio-focus](https://developer.android.com/media/optimize/audio-focus)



### Our work

Prototype code and measurements  
[github.com/prasadxa/itantra](https://github.com/prasadxa/itantra)

Project website: overview, results, APK  
[itantra-106.pages.dev](https://itantra-106.pages.dev)

Demo film, 1 min 55 s, captioned  
[itantra-106.pages.dev/demo](https://itantra-106.pages.dev/demo)



### Website

[overview](#) · [results](#) · [APK](#)



### Demo film

1 min 55 s

Licences as stated on each model card. No cloud API is used for speech recognition or speech synthesis.